



Ephrance Eunice Namugenyi, PhD

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[LinkedIn Profile](#)

ACADEMIC PROFILE

Postdoctoral-level researcher and Lecturer with a decade of experience in Data Communications, Software Engineering, and Communication Networks. Research expertise lies at the intersection of wireless communications, IoT systems, edge intelligence, and adaptive network architectures. Demonstrated record of peer-reviewed publications and international fellowships.

EDUCATION

PhD in Software Engineering <i>Makerere University</i>	2021 - 2026
MSc in Data Communications & Software Engineering <i>Makerere University</i>	2015 - 2019
BSc in Telecommunications Engineering <i>Makerere University</i>	2008 - 2012

ACADEMIC & PROFESSIONAL APPOINTMENTS

Lecturer - Electrical & Electronics Engineering <i>Kyambogo University</i>	2013 - Present
- Courses: Communication Systems, Satellite Comm, Computing for Engineers. - Supervision: Wireless networks, IoT systems, and embedded systems.	
Technical Support Staff - Embedded Systems <i>CRTT, Makerere University</i>	2012 - 2013
- Developed embedded software for lighting systems of Uganda's first solar bus (Kayoola Bus). - Technologies: C, C++, MATLAB, PXROS, and Python in real-time systems.	
User Experience Developer - ARMS Project <i>Makerere University</i>	2010 - 2012
- Designed UI and developed SMS-based Results Request System for low-connectivity environments.	

FELLOWSHIPS, GRANTS & SELECTED PROJECTS

- * STEP Fellowship, ICTP (Italy) - PhD Research Attachment (2022-2024)
- * IEEE SIGHT COVID-19 Innovations - Team Lead (USD 5,000 Grant for healthcare sensing)
- * 3MT People's Choice Award (Environment) - Washington University in St. Louis (2023)
- * Vital Signs Monitoring & Energy-Aware Network Modeling (NS-3 & MATLAB)
- * Award of Excellence (40 Days 40 Fintechs) & IEEE SIGHT Honorable Mention (2021)

PUBLICATIONS & RESEARCH

GOOGLE SCHOLAR: <https://scholar.google.com/citations?user=ZbMeRqEAAAAJ>

- * Namugenyi et al. (2025). Adaptive Network Switching Algorithm. *Frontiers in IoT*.
- * Namugenyi et al. (2025). Energy Consumption Optimization in LoRa Networks. *Springer*.
- * Namugenyi et al. (2024). Robust Sensor Network Design. *SVC Vancouver*.

TECHNICAL SKILLS

Programming: Python, C, C++ | ML: Machine Learning for Network Optimization
Modeling: NS-3, MATLAB | Networking: IoT, LPWAN, Wi-Fi, GSM/LTE, Edge Computing